

Clement (Kai Hei) Chu
Curriculum Vitae
clement[dot]chu[at]mail[dot]utoronto.ca

EDUCATION

University of Toronto

Mathematics and Its Applications Specialist in Probability and Statistics

Completion Year: May 2027

Relevant Coursework: Real Analysis, Complex Variables, Theory of Statistics, Theory of Statistical Inference, Statistical Computation, Time Series, Regression Analysis, Group Theory, Sampling Theory, Nonlinear Optimization Theory

RESEARCH INTERESTS

Probability theory, Generative modeling, Score-based and diffusion methods.

PUBLICATIONS

C. K. H. Chu. *Preservation of Baire Spaces under Homeomorphisms.* TRUNK Undergraduate Research Journal, 2025. [PDF]

EXPOSITORY PAPERS AND PROJECTS

Spectral Graph Theory (Directed Reading Project)

Department of Mathematics, University of Toronto

Jan–May 2025

Supervisor: Narmada Varadarajan (PhD)

- Studied Perron-Frobenius theory, expander graphs, and pseudorandom graph constructions.
- Analyzed spectral properties of graph Laplacians and their applications to graph embeddings. [Slides]

Causal Inference (Directed Reading Project)

Department of Statistical Sciences, University of Toronto

Jan–May 2026

Supervisor: Arian Hashmzadeh (PhD)

Expository Paper: Analysis of Robustness of Linear Estimators via ℓ^p and Dual Norm Lipschitz Bounds

Informally guided by PhD student *Jason (Sin Hang) Yeung*. [PDF]

Are Improper Priors Objective in Statistical Inference?

Course project for STA422/2162 under Prof. Michael Evans

2026

- Investigated the objectivity of improper priors, such as default and reference priors, with theoretical and philosophical perspectives. [PDF]

“A Note in Probability and Statistics”

2025–Present

- Developing expository notes on probability theory and statistical inference for early undergraduate audiences.

TEACHING EXPERIENCE

Private high school tutor in Mathematics

2023–Present

- Tutored high school students who are preparing for Cambridge IGCSE and IB public exams.

ADDITIONAL INFORMATION

Programming and Typesetting Languages: Python, R; L^AT_EX

Libraries: ggplot2, Quarto, NumPy

Awards: Dean’s List Scholar (2024–2025)

Languages: English (Native), Cantonese (Native), Mandarin (Native), Japanese (Intermediate)